Jenkins

Introduction to Jenkins

What is Jenkins?

- Jenkins is an **open-source automation server** used to automate software development processes.
- It helps implement **Continuous Integration (CI)** and **Continuous Delivery (CD)**.
- Written in **Java** and supports plugins for integrating various tools.

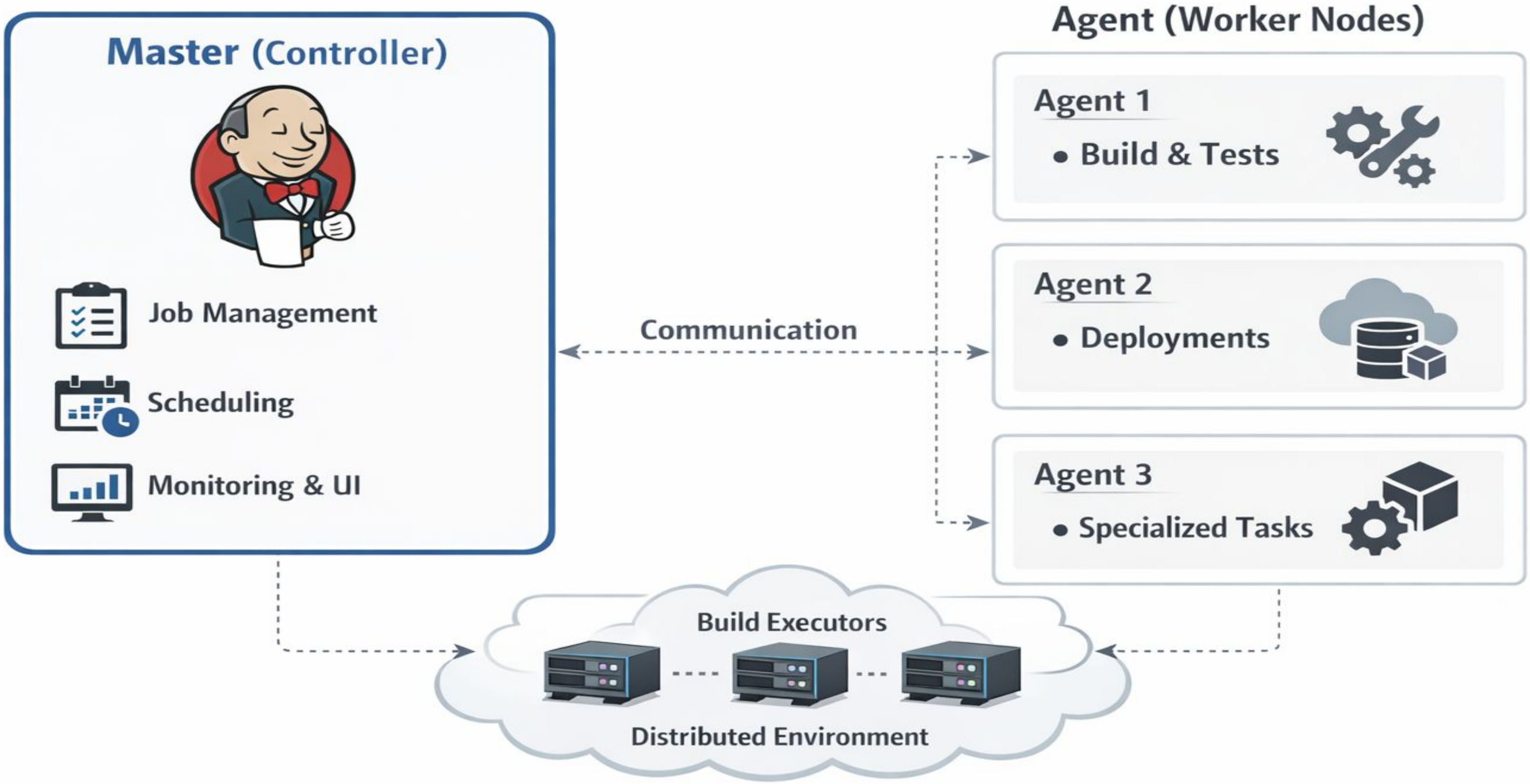
Why Jenkins?

- Automates repetitive tasks like build, test, and deployment
- Reduces human errors
- Speeds up software delivery
- Provides real-time feedback on code changes

Key concept

Developers push code → Jenkins automatically builds, tests, and deploys it.

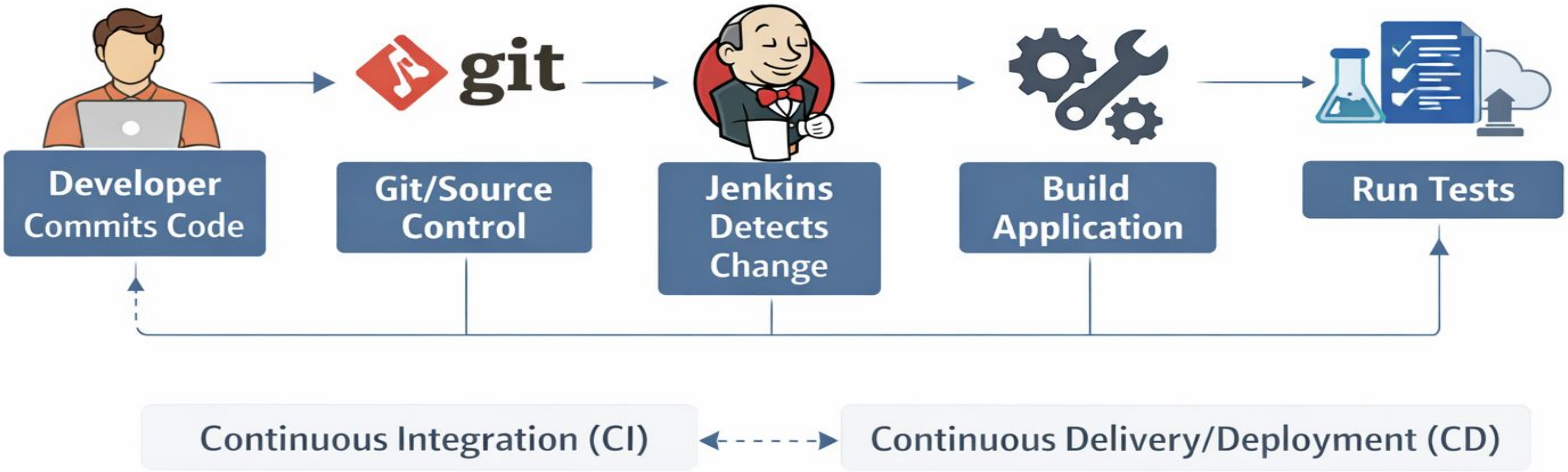
Jenkins Architecture



Key Features of Jenkins

- **Extensible:** 1000+ plugins available
- **Platform Independent:** Works on Windows, Linux, macOS
- **Easy Integration:** Works with Git, Docker, Maven, Kubernetes, etc.
- **Distributed Builds:** Supports master-agent architecture
- **Pipeline as Code:** Define workflows using `Jenkinsfile`

Jenkins Pipeline Flow



Pipeline Types

- Declarative Pipeline (simple & structured)
- Scripted Pipeline (flexible & advanced)

Key Differences(Quick Comparison)

Feature	Declarative Pipeline	Scripted Pipeline
Syntax	Structured & predefined	Flexible (Groovy-based)
Ease of Use	Easy (beginner-friendly)	Moderate to difficult
Flexibility	Limited	Very high
Readability	High	Lower
Error Handling	Built-in	Manual
Use Case	Standard CI/CD	Complex workflows

Use Cases

- Continuous Integration (CI) → automatic build & test on every commit
- Continuous Delivery/Deployment (CD) → deploy applications automatically
- Automating testing (unit, integration tests)
- Building Docker images & deploying to cloud
- Scheduled jobs (cron-based automation)

Real-World Flow

- Developer commits code to Git
- Jenkins detects change
- Builds the application
- Runs tests
- Deploys to staging/production